

Trustworthy Recommendation System

Retrieval, Ranking, and Trustworthy Offline Evaluation

TEAM MEMBERS

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RANKING AT SCALE

How do we retrieve and rank relevant items quickly from a very large catalog?

OFFLINE EVALUATION

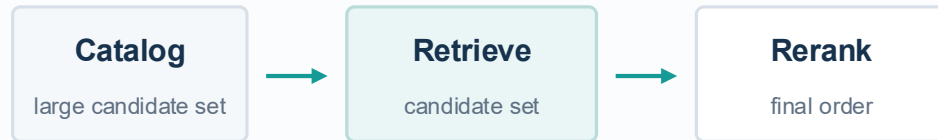
Can we estimate whether a new ranker is better before deciding what is worth testing live?

Recommendation systems have to solve two linked problems

1

Ranking at scale

Users need relevant results quickly from catalogs that are too large to score exhaustively. Ranking therefore has to happen in stages under a hard latency budget.



2

Trustworthy offline evaluation

Logged interaction data is biased by the model that produced it: users only interacted with what they were shown, and top positions receive more attention regardless of relevance.



PROJECT QUESTION

Can we estimate a new ranker's performance from biased logs accurately enough to decide what's worth testing live?

A consumer-facing recommendation experience that can be refined

USER WORKFLOW

- 1 Establish or express preferences**
The user indicates what they are looking for.
- 2 Receive a ranked set**
The system returns a ranked set of recommendations.
- 3 Refine preferences**
The user adjusts constraints when the first results are not satisfactory.
- 4 Receive updated recommendations**
The ranking changes in response to the revised intent.

VALUE PROPOSITION

Reduce the effort required to find relevant items in a large catalog by combining long-term behavioral signals with preferences expressed during the current interaction.

PLANNED INTERACTION LAYER

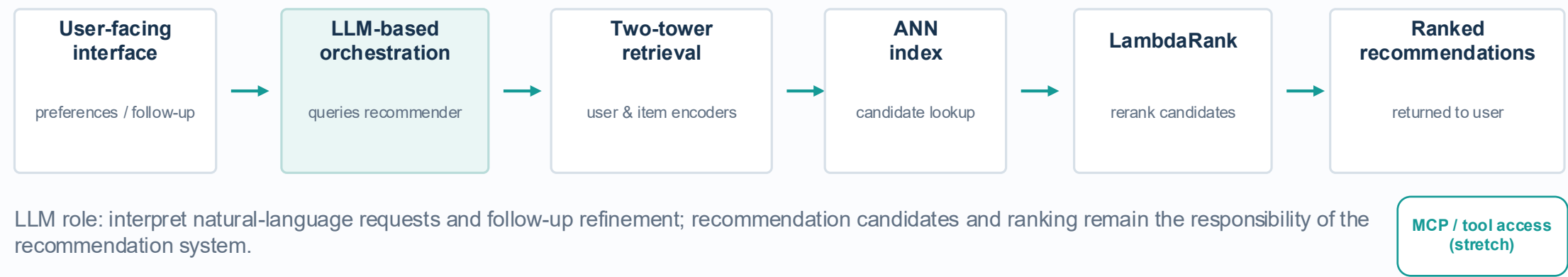
The user-facing interface will support natural-language requests and follow-up refinement through an LLM-based orchestration layer that queries the recommendation system rather than generating recommendations directly.

Domain selection remains open until the initial dataset audit: MovieLens and Amazon Reviews 2023 are the possibilities

Proposed architecture

Current design direction based only on the proposal.

ONLINE RECOMMENDATION FLOW



LLM role: interpret natural-language requests and follow-up refinement; recommendation candidates and ranking remain the responsibility of the recommendation system.

OFFLINE POLICY EVALUATION



Position bias and propensity correction will use the information available in the selected evaluation dataset.

Data selection and success criteria

RECOMMENDATION DATA	OFFLINE TECHNICAL METRICS	PRODUCT & RESPONSIBLE EVALUATION
MovieLens or Amazon Reviews 2023 Select after auditing user-history depth, item metadata, timestamps, labels, and licensing.	Retrieval Recall@k + recall/latency Pareto Target: approx. Recall@100 ≥ 0.80; p99 ANN < 50 ms	User scenarios Predefined scenarios test whether stated preferences and constraints are reflected in recommendations.
OPE DATA	Ranking NDCG@10 + MRR Against BM25-style + popularity baselines on a temporal split	Popularity bias Where supported by the data, examine recommendation concentration and whether high-popularity items are disproportionately favored.
Open Bandit Dataset Leading OPE dataset because it contains real logged bandit feedback with known propensities. Separate recommendation and OPE datasets are acceptable if needed.	OPE Error + variance + CI coverage Reported by log size and policy overlap	LLM failure testing If the LLM interaction layer is used, hallucinated or unsupported outputs are included in failure testing.
<i>Licensing and usage constraints will be documented before implementation.</i>	End-to-end Serving latency Request to ranked results	

Scope and 14-week delivery plan

CORE SCOPE

Two-tower retrieval • ANN candidate retrieval • LambdaRank reranking • recommendation baselines • consumer-facing interface • natural-language refinement • reproducible training/evaluation • DM/IPS/SNIPS/DR • OPE reliability • end-to-end user-flow testing

STRETCH

MCP-based tool access to the recommendation/search backend

OUT OF SCOPE

Production-scale deployment • powered live A/B test • general-purpose multi-domain platform • LLM as recommendation/ranking model

14-WEEK ROADMAP

1–2	3–5	6–7	8–9	10–12	13–14
Dataset / domain	Two-tower	ANN + UI	Rerank + E2E	OPE + validate	Harden + present
audit • licensing • temporal split • baselines • UI mockup	sampled softmax • logQ	latency • hard-negative mining • interface prototype	LambdaRank • cross features • integration	DM/IPS/SNIPS/DR • interface • overlap • system evaluation	failure analysis • packaging • report • demo • showcase

MAIN RISKS

Dataset suitability

IPS variance

14-week scope

Mitigation: use separate recommendation/OPE datasets if needed; reduce reranker sophistication and advanced features.